

# **Quantifying Densification versus Shear Flow during Glass Nanoindentation via Atomic-Scale Descriptors**

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## Abstract

Plastic deformation in glasses under sharp contact loading can occur through distinct mechanisms, ranging from volumetric densification to shear flow. While molecular dynamics simulations provide detailed atomic-scale information, a transferable framework to quantitatively compare these mechanisms across chemically distinct glasses is still lacking. In this work, we introduce a set of atomic-scale descriptors to quantify densification and shear flow during nanoindentation, derived from local irreversible deformation fields obtained from molecular dynamics simulations. The analysis combines volumetric and deviatoric strain components, atomic-scale densification, and the non-affine displacement measure  $D^2_{\min}$ , with particular emphasis on the residual deformation after unloading to isolate plastic effects.

The proposed approach is applied to vitreous  $\text{SiO}_2$ , a sodium silicate, albite, anorthite glasses and the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass, spanning densification-dominated, shear-dominated, and mixed deformation behaviours. We show that strain-based descriptors alone are insufficient to uniquely discriminate these mechanisms. To address this limitation, we introduce an end-member-based Densification–Shear Fraction (DSF), constructed from residual densification and non-affine rearrangement intensity. The DSF provides a continuous and physically grounded quantification of the relative contribution of densification and shear flow. This framework establishes a unified and transferable atomic-scale description of nanoindentation-induced plasticity, applicable to chemically diverse glassy systems.

## 1. Introduction

Nanoindentation is a primary route to quantify the small-scale mechanical response of glasses and to connect composition–structure to hardness, damage resistance, and contact-induced cracking.[1–7] Unlike crystalline solids, however, the inelastic response of oxide glasses under sharp contact typically arises from two intertwined mechanisms: permanent densification (volume compaction) and shear flow (isochoric plastic rearrangements).[7–14] Their relative contribution controls the shape of the residual imprint (sink-in vs pile-up), the spatial distribution of residual strain, and ultimately the propensity for crack initiation and propagation. Yet, separating and quantifying these two mechanisms in a manner that is comparable across glass families remains challenging, especially when only load–displacement data are available.

A large body of work has established that fused silica represents a paradigmatic case of densification-dominated plasticity under indentation, whereas modified silicate and aluminosilicate glasses exhibit mixed responses in which shear flow becomes increasingly relevant as network connectivity decreases and modifier content increases.[7,15] Spatially resolved experimental techniques have been instrumental in elucidating these mechanisms. In particular, Raman micro-spectroscopy has enabled direct mapping of indentation-induced permanent densification in silica with micrometric resolution, revealing strong densification gradients beneath the indenter and associated densification-induced hardening effects.[16,17] Similar approaches have demonstrated that, in modifier-rich silicate glasses, densification is progressively accompanied by shear-driven deformation, leading to broader plastic zones and reduced volumetric compaction.[18]

Continuum modeling and constitutive descriptions have further framed densification as a pressure-dependent plastic process that competes with shear flow, emphasizing that the balance between the two mechanisms is strongly composition-dependent and directly impacts indentation-induced cracking and damage resistance.[11,19] These studies highlight that volumetric strain alone is insufficient to describe the plastic response of complex silicate networks, where shear localization and densification frequently coexist.

In contrast, metallic glasses deform almost exclusively through shear-driven mechanisms. Their plasticity is governed by the activation and collective organization of shear transformation zones (STZs), which produce highly non-affine atomic rearrangements with minimal net densification.[20,21] Under indentation, metallic glasses typically exhibit extended shear-dominated plastic zones, pronounced pile-up, and limited volumetric strain, in sharp contrast with densification-prone oxide glasses.[22] This fundamental difference in

deformation mechanisms makes metallic glasses a natural shear-dominated end member for comparative analyses of indentation-induced plasticity.

Despite these advances, quantitatively disentangling densification and shear flow during nanoindentation remains challenging. Experimental indentation measurements primarily provide global load–displacement curves, from which hardness and modulus can be extracted, but do not uniquely resolve the underlying plastic deformation mechanisms. Even when complemented by spatially resolved probes such as Raman spectroscopy, the analysis is typically restricted to oxide glasses and relies on indirect indicators of densification, making direct comparisons across chemically distinct glass families difficult. Atomistic simulations offer a powerful complementary perspective by providing direct access to local deformation fields at the atomic scale. Molecular dynamics nanoindentation simulations have been extensively employed to investigate plastic deformation in both oxide and metallic glasses, revealing the spatial distribution of strain, structural rearrangements, and shear localization beneath the indenter.[17,23–28] In particular, the volumetric and deviatoric strain components and the non-affine displacement measure  $D^2_{\min}$  have emerged as robust metrics to identify shear transformation zones and shear-dominated plastic events in amorphous materials.[21]

Importantly, such indicators are, in principle, well defined and applicable across different glass families, including oxide and metallic glasses. However, each of them captures only a specific aspect of the deformation process. Local strain measures quantify the magnitude of affine volumetric or deviatoric deformation, while  $D^2_{\min}$  selectively probes non-affine rearrangements associated with shear-driven plasticity. As a consequence, interpreting the relative contribution of densification and shear flow based on any single indicator remains non-trivial, particularly when comparing glasses that exhibit mixed deformation mechanisms.

This limitation becomes evident when attempting to establish a direct, quantitative comparison between densification-dominated glasses, such as vitreous silica, mixed-mechanism silicate compositions, and shear-dominated metallic glasses. While individual indicators can clearly highlight qualitative differences—for instance, localized densification beneath the indenter in silica or extended non-affine activity in metallic glasses—they do not, by themselves, provide a normalized measure of the relative weight of densification and shear that is transferable across chemically and structurally distinct systems.

Here we address this challenge by building upon these established atomistic indicators rather than replacing them. By combining atomic-scale densification and non-affine

rearrangement metrics within a unified representation, we construct a descriptor space that explicitly captures the competition between volumetric and shear-driven plasticity. This approach allows the relative contribution of densification and shear flow to be quantified in a continuous and physically interpretable manner, enabling direct comparison across oxide and metallic glasses within a single framework.

Here, we demonstrate the relevance and robustness of the proposed descriptors by applying them to a representative set of glassy systems: vitreous silica, a  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass (denoted CuZr hereafter), respectively; a sodium silicate glass with composition  $12.5\%\text{Na}_2\text{O}\cdot 87.5\%\text{SiO}_2$  (denoted NS), the albite glass ( $12.5\%\text{Na}_2\text{O}\cdot 12.5\%\text{Al}_2\text{O}_3\cdot 75\%\text{SiO}_2$ , denoted AL), and the anorthite glass ( $25\%\text{CaO}\cdot 25\%\text{Al}_2\text{O}_3\cdot 50\%\text{SiO}_2$ , denoted AN). These materials span the full spectrum of deformation behavior under indentation, from densification-dominated ( $v\text{-SiO}_2$ ) to shear-dominated plasticity ( $\text{Cu}_{64}\text{Zr}_{36}$ ), with sodium- and calcium-bearing aluminosilicate glasses exhibiting intermediate and mixed responses. By analysing configurations before indentation, at maximum penetration, and after unloading, we explicitly isolate the irreversible contribution to deformation and its spatial distribution.

The paper is organized as follows. We first describe the computational methodology, including glass preparation, the nanoindentation protocol, and the atomistic framework used to characterize local deformation. In particular, we introduce the definitions of atomic-scale densification, non-affine rearrangements, and local volumetric and deviatoric strain measures, together with the criteria adopted to identify plastically active atoms. We then present a series of scalar and dimensionless descriptors derived from these atomistic fields to quantify densification- and shear-related contributions to plastic deformation.

Subsequently, the Results section begins with an analysis of the macroscopic nanoindentation response in terms of load–displacement curves and hardness, providing a validation of the simulation protocol and highlighting the limitations of global mechanical metrics for identifying deformation mechanisms. We then examine atomic-scale maps of residual densification, non-affine rearrangements, and strain fields after unloading, revealing qualitative differences in the spatial extent and localization of plastic deformation across glass compositions. Building on these observations, we assess the performance of the proposed strain-based and mechanism-resolving descriptors. Finally, we introduce an end-member-based Densification–Shear Fraction (DSF), which provides a unified and transferable quantification of the relative contribution of densification and shear flow across chemically distinct glasses, bridging oxide and metallic systems within a single, physically grounded framework.

## 2. Computational Methods.

### 2.1 Glass Preparation.

The glass structures were generated using the traditional melt-quench approach[29] via molecular dynamics simulations, using the LAMMPS code[30]. For the silicate glasses ( $\text{SiO}_2$ , NS, AL and AN glasses) we started by randomly placing the atoms in a cubic box whose side is related to the experimental density as reported in **Table 1**. The Table also contains the number of each atomic species, total number of atoms and box size.

Then, the system is melted at 3000K and kept at this temperature for 40 ps using the NVT ensemble and then rapidly cooled down at 300K by applying a monotone cooling rate of 5 K/ps using the NPT ensemble (Nosé–Hoover)[31] with atmospheric pressure. Finally, the box is equilibrated at 300K for 200ps. The equation of motion was solved using the Verlet-Leapfrog algorithm with a time step of 0.002 ps. The BMP interatomic potential,[32] whose functional form is reported in the ESI, has been used to describe atomic interactions because it reproduces well the structure and elastic properties of multicomponent oxide glasses.[33,34] The short-range interactions were evaluated using a cut-off of 7Å whereas the long-range Coulombic interactions have been evaluated using the Ewald summation method with cut-off distances of 12Å.

The initial structure of the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass structure consisted of a cubic simulation box containing 62 500 atoms with a Cu:Zr atomic ratio of 64:36, corresponding to a box length of approximately 100 Å after equilibration at ambient pressure. Atomic positions were initially assigned on a face-centered cubic lattice and randomly occupied by Cu and Zr atoms according to the target composition to ensure a homogeneous starting configuration.

Interatomic interactions in the Cu–Zr system were described using an embedded-atom method potential of the Finnis–Sinclair type developed by Mendelev *et al.*[35,36]

The system was first heated to 2000 K, well above the liquidus temperature of the alloy, and equilibrated for 100 ps in the NPT ensemble using a Nosé–Hoover thermostat and barostat at zero external pressure to ensure complete melting and loss of memory of the initial configuration. Subsequently, the liquid was cooled down to 300 K using a linear cooling rate of 5 K  $\text{ps}^{-1}$  under NPT conditions at atmospheric pressure, consistent with typical cooling rates employed in molecular-dynamics studies of metallic glass formation.[37–39] Finally, the resulting amorphous structure was equilibrated at 300 K for an additional 100 ps in the NPT ensemble to obtain a stress-free bulk metallic glass. The equations of motion were integrated using the velocity-Verlet algorithm with a time step of 0.002 ps. The final

equilibrated structure exhibited a density of approximately  $7.5 \text{ g cm}^{-3}$ , in good agreement with reported experimental and simulation values for  $\text{Cu}_{64}\text{Zr}_{36}$  bulk metallic glasses. [37–39]

Table 1: Glass compositions: percentage of oxides, number of atoms, density of final boxes

|                               | Na or Ca<br>atoms | Al atoms | Si atoms | O atoms | Tot<br>atoms | $\rho \text{ (g*cm}^{-3}\text{)}$ | $L_{\text{box}} \text{ (Å)}$ |
|-------------------------------|-------------------|----------|----------|---------|--------------|-----------------------------------|------------------------------|
| <b><i>SiO<sub>2</sub></i></b> |                   |          | 27000    | 54000   | 81000        | 2.200                             | 106.974                      |
| <b><i>AN</i></b>              | 7424              | 14848    | 14848    | 59392   | 96512        | 2.478                             | 111.442                      |
| <b><i>AL</i></b>              | 6152              | 6152     | 18464    | 49232   | 80000        | 2.320                             | 102.865                      |
| <b><i>NS</i></b>              | 5920              | -        | 23712    | 50384   | 80016        | 2.299                             | 104.700                      |
| <b><i>CuZr</i></b>            | -                 | -        | -        | -       | 62500        | 7.50                              | 100.360                      |

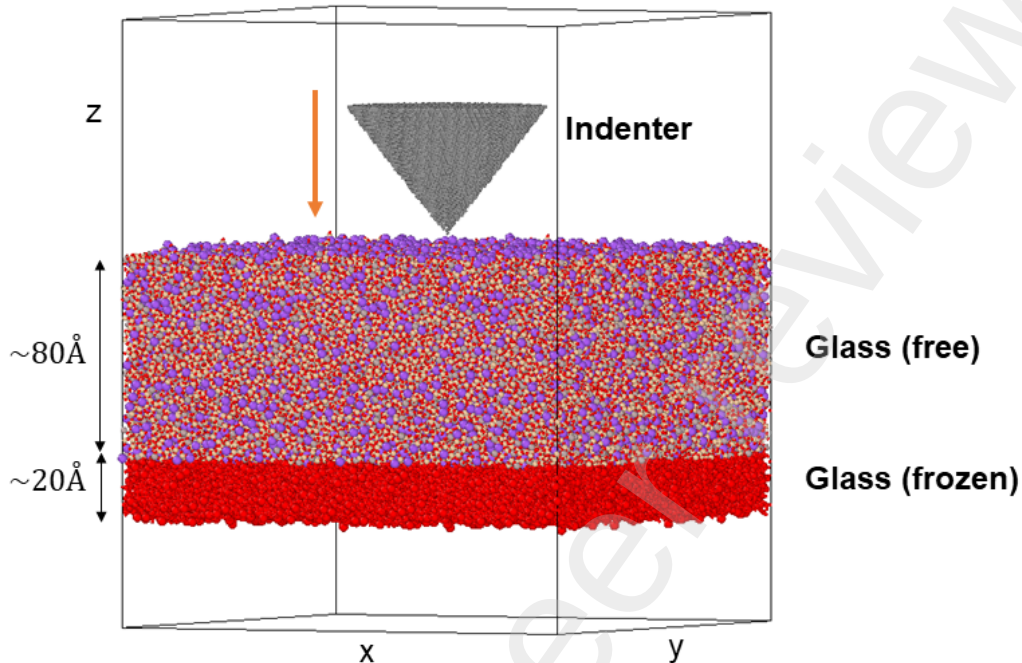
## 2.2 Nanoindentation Protocol

From the bulk configurations, we generated two free surfaces by extending the simulation cell along the z-axis to introduce a vacuum region large enough to prevent interactions between periodic images. The resulting slab models were then replicated twice along both the x and y directions to obtain larger surfaces of approximately 200 Å lateral size, containing between 250k and ~386k atoms. These extended slabs were subsequently relaxed and reconstructed through a heating–cooling equilibration cycle in the NVT ensemble to remove residual stresses and allow surface reconstruction. The system was first heated up from 300K to 1600K with a heating rate of 5K/ps and then cooled down from 1600K to 300K with the same rate, followed by equilibration at 300K for 400ps.

After a structural analysis of the pristine and pre-densified surfaces, we performed the indentation test and analysed the results during the simulation path.

The nanoindenter tip was modelled as a conical body composed of carbon atoms arranged in a diamond lattice. The cone axis was oriented along the z-direction, with the apex positioned above the slab surface. The cone geometry was chosen to reproduce the equivalent contact area of a standard Berkovich pyramidal tip.[40] Following the classical conversion between the three-sided Berkovich geometry ( $65.3^\circ$  face angle,  $142.3^\circ$  total included angle) and an axisymmetric cone providing the same projected contact area for a given penetration depth, the corresponding half-apex angle is  $37.3^\circ$ , equivalent to a total included angle of  $74.6^\circ$ . For the selected base radius of 40 Å, this angle yields a cone height of approximately 52.5 Å. The final tip model contained 15811 carbon atoms. The use of a conical model instead of the exact three-sided Berkovich pyramid allows one to preserve the correct contact area–depth relationship while ensuring axial symmetry, thus significantly

reducing computational cost and facilitating the analysis of stress and strain fields beneath the indenter. The Typical Nano-Indentation Model with slab and conical tip is shown in **Figure 1**.



**Figure 1** – Schematic representation of the nanoindentation set up

The interactions between carbon atoms in the indenter were described using the Tersoff bond-order potential in the 1994 formulation of Tersoff,[41] as implemented in LAMMPS. Finally, the tip-glass interactions have been modelled by using non-bonding Buckingham potential with  $A=1000$  eV,  $\rho=0.25$  Å and  $C = 0$  eV/Å<sup>6</sup>.

This functional form was selected to prevent unphysical adhesion or charge transfer between the carbon tip and the glass surface, ensuring that the interaction is dominated by short-range Pauli repulsion upon contact. The chosen parameters produce a steep repulsive wall that effectively mimics the mechanical constraint of a hard, chemically inert diamond tip while avoiding unrealistic penetration or bonding to surface atoms. Similar purely repulsive potentials have been widely employed in nanoindentation simulations to approximate hard–soft material contacts where chemical interactions are negligible and the main effect of the indenter is to impose a local mechanical load.[25,26]

The indentation was simulated in NVT ensemble at a constant temperature of 300K lowering the tip along the z-axis with a rate of 0.5Å/ps until the entire 5nm tip penetrated the glass surface. Before unloading the tip was held at the maximum penetration for 100ps. Finally,

the tip has been unloaded with the same velocity. During the simulations the atoms in the bottom layer (20Å) of the glass have been fixed.

Comparable loading rates (0.1–0.5 Å ps<sup>-1</sup>) have been widely adopted in previous molecular-dynamics nanoindentation studies to balance physical realism and simulation efficiency.[23,26–28]

Hardness values have been extracted from the load-penetration depth curves following the Oliver and Pharr method.[42] Hardness is defined as the ratio between the maximum load ( $P_{\max}$ ) over the *contact* area:

$$H = \frac{P_{\max}}{A_c} \quad (1)$$

Here,  $A_c$  is the projected contact area, which for a conical indenter with apex angle  $\varphi$  is computed as:

$$A_c = 4h_c^2 \tan^2(\varphi/2) \quad (2)$$

where  $h_c$  is the contact depth which is computed by the formula:

$$h_c = h_{\max} - \varepsilon \frac{P_{\max}}{S} \quad (3)$$

with  $h_{\max}$  being the maximum penetration depth,  $\varepsilon$  a constant which depends on the geometry of the indenter ( $\varepsilon = 0.72$  for a conical indenter), and  $S$  is the contact stiffness computed as the derivative of the loading  $P$  with respect to the penetration depth  $h$  at the maximum loading  $(dP/dh)_{P_{\max}}$ . The contact stiffness is computed by fitting the unloading curves with the functional form:

$$P = a(h - h_f)^m \quad (4)$$

Where  $h_f$  is the depth of plastic deformation whereas  $a$  and  $m$  are the fitting parameters.

### 3. Definition of Densification-Shear Descriptors.

#### 3.1 Atomic Strains Calculation.

To characterize atomic-scale deformation during nanoindentation, we computed a local deformation gradient  $\mathbf{F}_i$  for each atom  $i$  of the glass network, following the framework originally introduced by Falk and Langer[21] and already used in the analysis of amorphous plasticity.[19] All quantities were evaluated relative to an undeformed reference configuration corresponding to the initial slab prior to indentation. Carbon atoms belonging to the indenter were excluded from the analysis.

For a given atom  $i$ , a set of neighbouring atoms  $j$  was identified within a cutoff radius  $r_c$  in the reference configuration. The neighbour vectors in the reference and deformed configurations were defined as  $\mathbf{r}_{ij}^0 = \mathbf{r}_j^0 - \mathbf{r}_i^0$  and  $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$ , respectively.  $\mathbf{r}_k^0$  and  $\mathbf{r}_k$  denote

the atomic positions before and after deformation, respectively. Periodic boundary conditions were applied using the minimum-image convention when constructing these vectors. The local deformation gradient  $\mathbf{F}_i$  was obtained by minimizing, in a least-squares sense, the difference between the deformed neighbour vectors and those predicted by an affine transformation of the reference vectors:

$$\min_{\mathbf{F}_i} \sum_{j \in N_i} \|\mathbf{r}_{ij} - \mathbf{F}_i \mathbf{r}_{ij}^0\|^2 \quad (5)$$

The solution of this minimization problem yields

$$\mathbf{F}_i = \mathbf{X}_i \mathbf{Y}_i^{-1} \quad (6)$$

With

$$\mathbf{X}_i = \sum_{j \in N_i} \mathbf{r}_{ij} \otimes \mathbf{r}_{ij}^0 \quad (7)$$

$$\mathbf{Y}_i = \sum_{j \in N_i} \mathbf{r}_{ij}^0 \otimes \mathbf{r}_{ij}^0 \quad (8)$$

Only atoms with a sufficient number of neighbours and a well-conditioned  $\mathbf{Y}_i$  tensor were retained to ensure numerical stability of the local fit.

From the deformation gradient, we computed the local strain tensor as:

$$\mathbf{E}_i = \frac{1}{2}(\mathbf{F}_i^T \mathbf{F}_i - \mathbf{I}) \approx \frac{1}{2}(\mathbf{F}_i + \mathbf{F}_i^T) - \mathbf{I} \quad (9)$$

where  $\mathbf{I}$  is the identity matrix. The off-diagonal components of  $\mathbf{E}_i$  (such as  $e_{yz}$ ) represent local shear strains, while the diagonal terms correspond to normal strains along each coordinate direction.

Therefore, the volumetric strain was defined as the trace of the strain tensor.

$$\varepsilon_{vol,i} = Tr(\mathbf{E}_i) \quad (10)$$

Providing a direct measure of local volume change (densification or dilation). The equivalent deviatoric strain was computed from deviatoric part of the strain tensor,

$$\mathbf{E}_i^{dev} = \mathbf{E}_i - \frac{1}{3}Tr(\mathbf{E}_i)\mathbf{I} \quad (11)$$

As

$$\varepsilon_{dev,eq,i} = \sqrt{\frac{2}{3} \mathbf{E}_i^{dev} : \mathbf{E}_i^{dev}} \quad (12)$$

This scalar quantity represents the magnitude of shear-like, volume-preserving deformation at the atomic scale and is analogous to von Mises strain in continuum mechanics.[43]

### 3.2 Non-Affine Displacement Descriptor.

In addition to the affine deformation captured by  $\mathbf{F}_i$ , we quantified local plastic rearrangements using the non-affine displacement measure  $D_{min}^2$ . For each atom  $i$ ,  $D_{min,i}^2$  is

defined as the minimum value of the same least-squares objective function used to determine  $\mathbf{F}_i$ :

$$D_{min,i}^2 = \sum_{j \in N_i} \|\mathbf{r}_{ij} - \mathbf{F}_i \mathbf{r}_{ij}^0\|^2 \quad (13)$$

Physically,  $D_{min,i}^2$  quantifies the part of the local atomic motion that cannot be described by a best-fit affine deformation and therefore serves as a sensitive indicator of irreversible, non-affine rearrangements. Small values of  $D_{min}^2$  correspond to elastic or nearly affine deformation, while large values indicate local plastic events such as shear transformation zones, bond switching or collapse of free volume.

### 3.3 Atomic-scale densification

In addition to non-affine rearrangements, we explicitly quantified the local volumetric contribution to plastic deformation through an atomic-scale measure of densification. For each atom  $i$ , the densification was computed from the determinant of the local deformation gradient  $\mathbf{F}_i$ .

Assuming local mass conservation within the neighbourhood, the relative change in atomic density is given by

$$\frac{\Delta \rho_i}{\rho_i} = \frac{1}{\det(\mathbf{F}_i)} - 1 \quad (14)$$

Where  $\det(\mathbf{F}_i)$  represents the local volume change associated with the affine deformation.

Positive values of  $\frac{\Delta \rho_i}{\rho_i}$  correspond to local densification, while negative values indicate local dilatation.

This definition provides a direct, dimensionless measure of volumetric plastic deformation at the atomic scale, fully consistent with the local strain tensor analysis. Unlike coordination-based or Voronoi-based measures, the present approach does not rely on an explicit definition of atomic volumes and remains applicable to chemically diverse systems, including both oxide and metallic glasses.

The atomic densification field was evaluated relative to the undeformed reference configuration and analysed primarily in the after-unload state, ensuring that the reported values reflect irreversible, plastic volume changes rather than elastic compression.

### 3.4 The descriptors

To quantitatively characterize the dominant plastic deformation mechanism under nanoindentation, we have introduced a set of complementary, dimensionless descriptors

derived from atomic-scale volumetric strain, deviatoric strain, densification, and non-affine displacement measures described above. All descriptors are computed from local, irreversible deformation fields evaluated with respect to the undeformed reference configuration.

First of all, we have to highlight that all descriptors are evaluated over a subset of atoms identified as *plastically active*. An atom  $i$  is classified as active if it undergoes a significant local irreversible strain, defined by the criterion:

$$\text{Active if } |\varepsilon_i^{vol}| > |\varepsilon_{th}^{vol}| \text{ or } |\varepsilon_i^{dev}| > |\varepsilon_{th}^{dev}|$$

A fixed absolute threshold  $\varepsilon_{th}^{vol} = \varepsilon_{th}^{dev} = \varepsilon_{th} = 0.1$  has been used in this work and the choice of the value was done on the basis of the strain maps reported in the result section.

### 3.4.1 Local model descriptors (intensity-based)

The following descriptors quantify the relative contribution of volumetric and deviatoric strain at the atomic scale, averaged over the active atoms.

**Descriptor  $I_1$ :** Mean Local Mode Ratio (MLMR)

$$I_1 = \left\langle \frac{|\varepsilon_i^{vol}|}{|\varepsilon_i^{vol}| + \varepsilon_i^{dev}} \right\rangle_{active} \quad (15)$$

This descriptor ranges from 0 (pure shear-dominated deformation) to 1 (pure densification-dominated deformation) and provides a continuous, local measure of the deformation mode.

**Descriptor  $I_2$ :** Fraction of Volumetrically dominated atoms

$$I_2 = \frac{N(|\varepsilon_i^{vol}| > \varepsilon_i^{dev})}{N_{active}} \quad (16)$$

Where  $N(|\varepsilon_i^{vol}| > \varepsilon_i^{dev})$  is the number of atoms for which  $|\varepsilon_i^{vol}| > \varepsilon_i^{dev}$ . The  $I_2$  descriptor measures the fraction of active atoms whose deformation is dominated by volumetric strain rather than shear. Unlike,  $I_1$ , this descriptor depends only on the relative ordering of strain components and is insensitive to their absolute magnitude.

**Descriptor  $I_3$ :** Signed mode index

$$I_3 = \left\langle \frac{|\varepsilon_i^{vol}| - \varepsilon_i^{dev}}{|\varepsilon_i^{vol}| + \varepsilon_i^{dev}} \right\rangle_{active} \quad (17)$$

$I_3$  ranges from +1 (densification-dominated) to -1 (shear-dominated) and provides a signed measure of the prevailing deformation mechanism.

**Descriptor  $I_4$ :** Mean Global Mode Ratio (MGMR)

The formula for the computation of this descriptor is similar to  $I_1$  but it is applied to the average atomic strains of the active atoms:

$$I_4 = \frac{\langle |\varepsilon_{vol}| \rangle}{\langle |\varepsilon_{vol}| \rangle + \langle |\varepsilon_{dev,eq}| \rangle} \quad (18)$$

By construction,  $I_4 \rightarrow 1$  corresponds to deformation dominated by volumetric strain (densification-dominated plasticity), whereas  $I_4 \rightarrow 0$  corresponds to deformation dominated by deviatoric strain (shear-dominated plasticity).

### 3.4.2 Mechanism-resolving descriptors

To overcome the limitations of purely intensity-based measures, we also introduce additional descriptors that explicitly quantify spatial extent and mechanistic coupling.

The first descriptor is the Extent Ratio (Y) defined as:

$$E = \frac{N(\varepsilon^{dev} > \varepsilon_{th}^{dev})}{N(|\varepsilon^{vol}| > \varepsilon_{th}^{vol})} \quad (19)$$

$$Y = \log_{10} E \quad (20)$$

This descriptor compares the spatial extent of shear-dominated deformation to that of densification. Large positive values of Y indicate extended shear flow with limited densification, whereas  $Y \sim 0$  corresponds to comparable extents of both mechanisms.

The Co-localization index (L)

$$L = \frac{N(|\varepsilon^{vol}| > \varepsilon_{th}^{vol} \cap \varepsilon^{dev} > \varepsilon_{th}^{dev})}{N(|\varepsilon^{vol}| > \varepsilon_{th}^{vol})} \quad (21)$$

Instead quantifies the degree of spatial overlap between densification and shear. Values close to unity indicate that shear and densification occur on the same atoms, whereas small values indicate decoupled mechanisms.

Finally, we quantify the coupling between densification and non-affine rearrangements using a rank-based correlation index defined as the Spearman correlation between densification and  $D_{min}^2$ :

$$C_S = Spearman(|\Delta\rho/\rho|, \log(D_{min}^2 + \epsilon)) \quad (22)$$

Where  $\epsilon$  is a small regularization constant. This descriptor is insensitive to absolute scales and directly probes whether irreversible rearrangements are accompanied by volumetric changes.

### 3.4.3 End-member-based Densification–Shear Fraction (DSF)

As will become evident from the results presented below, densification and non-affine rearrangements quantified by  $D_{min}^2$  emerge as the most effective and physically transparent descriptors for discriminating between different deformation components. This motivates the introduction of a combined scalar descriptor that explicitly couples these two measures,

providing a compact and directly comparable quantification of the relative contributions of densification and shear flow. To this end, we introduce an end-member-based Densification–Shear Fraction (DSF), constructed in a two-dimensional descriptor space spanned by residual densification and non-affine rearrangement intensity after unloading.

For each glass composition, we define a point

$$P = (S, D) = \left( \left\langle \log(D_{min}^2) \right\rangle_{active}, \langle \Delta\rho/\rho \rangle_{active} \right) \quad (23)$$

Where the averages are computed over plastically active atoms in the after-unload configuration. The coordinate D measures the average residual volumetric compaction and therefore quantifies densification-driven plasticity, while the coordinate S quantifies the intensity of non-affine rearrangements associated with shear-driven plastic deformation.

We selected vitreous SiO<sub>2</sub> and Cu<sub>64</sub>Zr<sub>36</sub> materials as end members assigning them the densification-dominate (large D and small S) and shear-dominated (large S and minimal densification) limiting deformation mechanisms, respectively.

These two materials define two fixed points in the (S, D) space

$$B = (S_{dens}, D_{dens}) \text{ of SiO}_2$$

$$A = (S_{shear}, D_{shear}) \text{ of Cu}_{64}\text{Zr}_{34}$$

To quantify how close a given glass is to each end member, we compute the distance between the point P and the two reference points A and B. Because the two axes  $\langle \log(D_{min}^2) \rangle$  and  $\langle \Delta\rho/\rho \rangle$  have different numerical ranges and physical units, distances are evaluated in a scaled descriptor space to prevent one coordinate from dominating the metric.

Specifically, we define a characteristic scale vector

$$\mathbf{s} = (s_S, s_D)$$

Where  $s_S$  and  $s_D$  are taken as the ranges of S and D spanned by the dataset (or, equivalently, the difference between the two end members). The scaled Euclidean distances to the shear- and densification-dominated end members are then defined as

$$d_{shear} = \sqrt{\left( \frac{S - S_{shear}}{s_S} \right)^2 + \left( \frac{D - D_{shear}}{s_D} \right)^2} \quad (24)$$

$$d_{dens} = \sqrt{\left( \frac{S - S_{dens}}{s_S} \right)^2 + \left( \frac{D - D_{dens}}{s_D} \right)^2} \quad (25)$$

This scaled distance ensures that deviations along the densification and rearrangement axes contribute comparably to the metric, reflecting their distinct but equally relevant physical roles.

Finally, the Densification–Shear Fraction is then defined as

$$f_{dens} = \frac{d_{shear}}{d_{shear} + d_{dens}} \quad (26)$$

$$f_{shear} = 1 - f_{dens} \quad (27)$$

By construction, this definition satisfies the following properties:

- $f_{dens}=1$  for the densification-dominated end member ( $\text{SiO}_2$ );
- $f_{dens}=0$  for the shear-dominated end member ( $\text{Cu}_{64}\text{Zr}_{36}$ );
- $0 < f_{dens} < 1$  for glasses exhibiting mixed deformation mechanisms.

Intermediate glass compositions are therefore interpreted as continuous mixtures of densification- and shear-driven plasticity, with  $f_{dens}$  providing a quantitative estimate of the relative contribution of densification.

## 4. Results.

### 4.1 Load vs Displacement Curves and Hardness

Figure 2 reports the simulated load–displacement curves obtained from nanoindentation of vitreous silica ( $\text{SiO}_2$ ), sodium silicate (NS), albite (AL), anorthite (AN), and the  $\text{Cu}_{64}\text{Zr}_{36}$  glasses. All systems exhibit a smooth loading branch up to a maximum indentation depth of approximately 46–47 Å, followed by elastic–plastic unloading with a finite residual depth  $h_f$ , indicating irreversible deformation. However, pronounced differences emerge in both the slope of the loading curves, the maximum load ( $P_{max}$ ) and the extent of elastic recovery, reflecting the distinct deformation mechanisms operative in silicate and metallic glasses.

Among the silicate glasses, the AN glass sustains the highest maximum load ( $P_{max} \approx 381 \text{ eV/Å}$ ) and exhibits the steepest loading curve, followed by AL and  $\text{SiO}_2$ . In contrast, the sodium silicate NS glass displays a significantly lower load response, consistent with the depolymerizing effect of Na modifiers on the silicate network.

Indeed, the  $\text{SiO}_2$ , AL, and AN systems have a fully polymerized three-dimensional networks in which all oxygen atoms are shared between network formers. This complete network connectivity results in a stiff mechanical response under indentation, as deformation requires collective rearrangement of a strongly constrained atomic framework. The enhanced load-bearing capacity of AN relative to the other tectosilicates can be directly attributed to its chemical composition. In anorthite, charge compensation of tetrahedrally coordinated Al is provided by  $\text{Ca}^{2+}$ , which forms stronger and shorter ionic bonds with oxygen compared to  $\text{Na}^+$ . The presence of Ca therefore promotes a more compact and mechanically robust network, increasing both packing efficiency and resistance to deformation under high contact pressures.

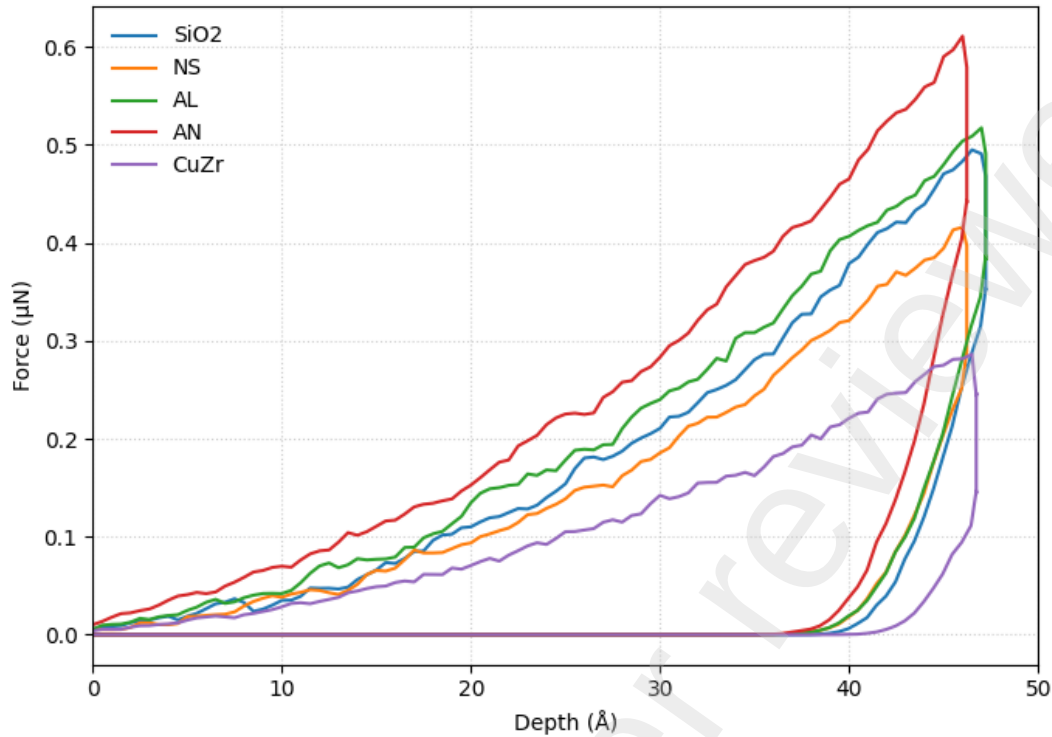
The metallic glass  $\text{Cu}_{64}\text{Zr}_{36}$  shows the lowest maximum load ( $P_{\text{max}} \approx 179 \text{ eV/\AA}$ ) for a comparable penetration depth, reflecting its softer response under sharp contact and the predominance of shear-driven plasticity.

Hardness values have been extracted from the simulated load–displacement curves using the Oliver–Pharr method and reported in Table 2. Vitreous  $\text{SiO}_2$  exhibits a high calculated hardness ( $H \approx 13.8 \text{ GPa}$ ), in reasonable agreement with experimental values reported for fused silica (between 9.5–12 GPa), the discrepancies probably due to inaccuracies on the interatomic potentials, the high strain rates and idealized indenter geometries. The NS glass show reduced hardness values of 11.9 GPa, respectively, also overestimating the experimental measurements of  $\sim 4.4\text{--}6 \text{ GPa}$  for sodium silicate glasses,[44] capturing the expected softening induced by network modifiers and mixed coordination environments. Albite and Anorthite exhibit higher simulated hardness among the silicate glasses ( $H \approx 14$  and  $17.2 \text{ GPa}$ , respectively), consistent with the highly polymerized and charge-balanced aluminosilicate framework, although direct experimental data for these glasses remain scarce. Sousa et al.[45] reported a value of 7.5 GPa for Albite glass whereas no data have been found for AN glass.

The  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass yields a calculated hardness of approximately 8.2 GPa, in close agreement with reported experimental values around 7–8 GPa obtained from micro- and nanoindentation studies[46] denoting the excellent accuracy of the interatomic potentials used.

The general agreement found for all the investigated systems supports the validity of the present simulation protocol across chemically diverse systems, despite intrinsic differences between silicate and metallic glasses.

Overall, the load–displacement curves and hardness values already indicate a clear mechanical separation between densification-prone silicate glasses and shear-dominated metallic glass behavior. However, hardness alone does not uniquely identify the underlying plastic deformation mechanisms, as materials with similar hardness may deform through fundamentally different combinations of densification and shear flow. This motivates the subsequent analysis of atomic-scale strain fields and non-affine rearrangements, aimed at disentangling the mechanistic origins of the observed macroscopic response.



**Figure 2.** Load vs Displacement curves for all investigated glasses during loading and unloading.

**Table 2:** Maximum load ( $P_{max}$ ), maximum indentation depth ( $h_{max}$ ), residual depth after unloading ( $h_f$ ), contact depth ( $h_c$ ), and hardness ( $H$ ) obtained from the Oliver–Pharr analysis for the investigated glasses.

|                        | $P_{max}$ (eV/Å) | $h_{max}$ (Å) | $h_f$ (Å) | $h_c$ (Å) | $H_{calc.}$ (GPa) | $H_{exp.}$ (GPa)   |
|------------------------|------------------|---------------|-----------|-----------|-------------------|--------------------|
| <b>SiO<sub>2</sub></b> | 308.9            | 46.5          | 39.0      | 42.2      | 13.8              | 9.5-12 GPa [47–49] |
| <b>NS</b>              | 259.6            | 46.0          | 36.5      | 41.7      | 11.9              | 4.4-6.0 [50]       |
| <b>AL</b>              | 323.0            | 47.0          | 36.0      | 42.9      | 14.0              | 7.5 [45]           |
| <b>AN</b>              | 381.3            | 46.0          | 35.5      | 42.0      | 17.2              | -                  |
| <b>CuZr</b>            | 178.7            | 46.5          | 40.5      | 41.6      | 8.2               | 7.2[46]            |

#### 4.2 Atomic-scale densification, non-affine rearrangements, and strain fields after unloading

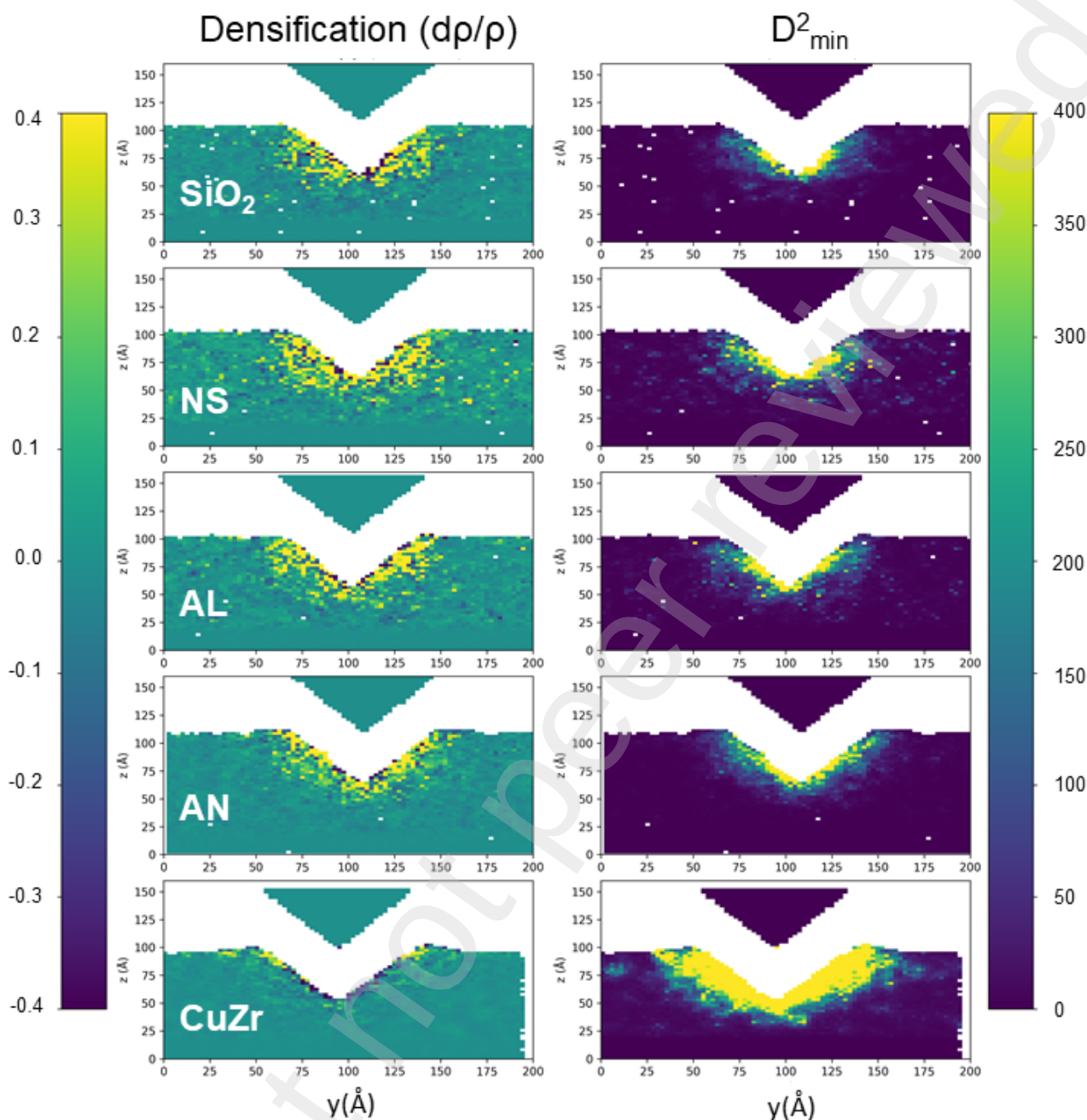
Figure 3 reports the atomic-scale maps of residual densification and non-affine rearrangements, while Figure 4 shows the corresponding volumetric and deviatoric strain fields, both evaluated in the *after-unload* configuration for all investigated glasses. These maps provide direct insight into the spatial distribution and extent of irreversible deformation beneath and around the indenter, highlighting clear qualitative differences between silicate and metallic glasses.

In fact, the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass exhibits a markedly different response compared to silicate systems. Its densification map shows only weak and localized volumetric changes, whereas the non-affine displacement measure ( $D^2_{\min}$ ) reveals an extended shear-dominated plastic zone spreading laterally well beyond the contact area and reaching the free surface. This broad distribution of high  $D^2_{\min}$  values is characteristic of shear-flow-dominated plasticity in metallic glasses and is accompanied by a pronounced pile-up around the indenter, consistent with the limited ability of the material to accommodate deformation through volumetric compaction.

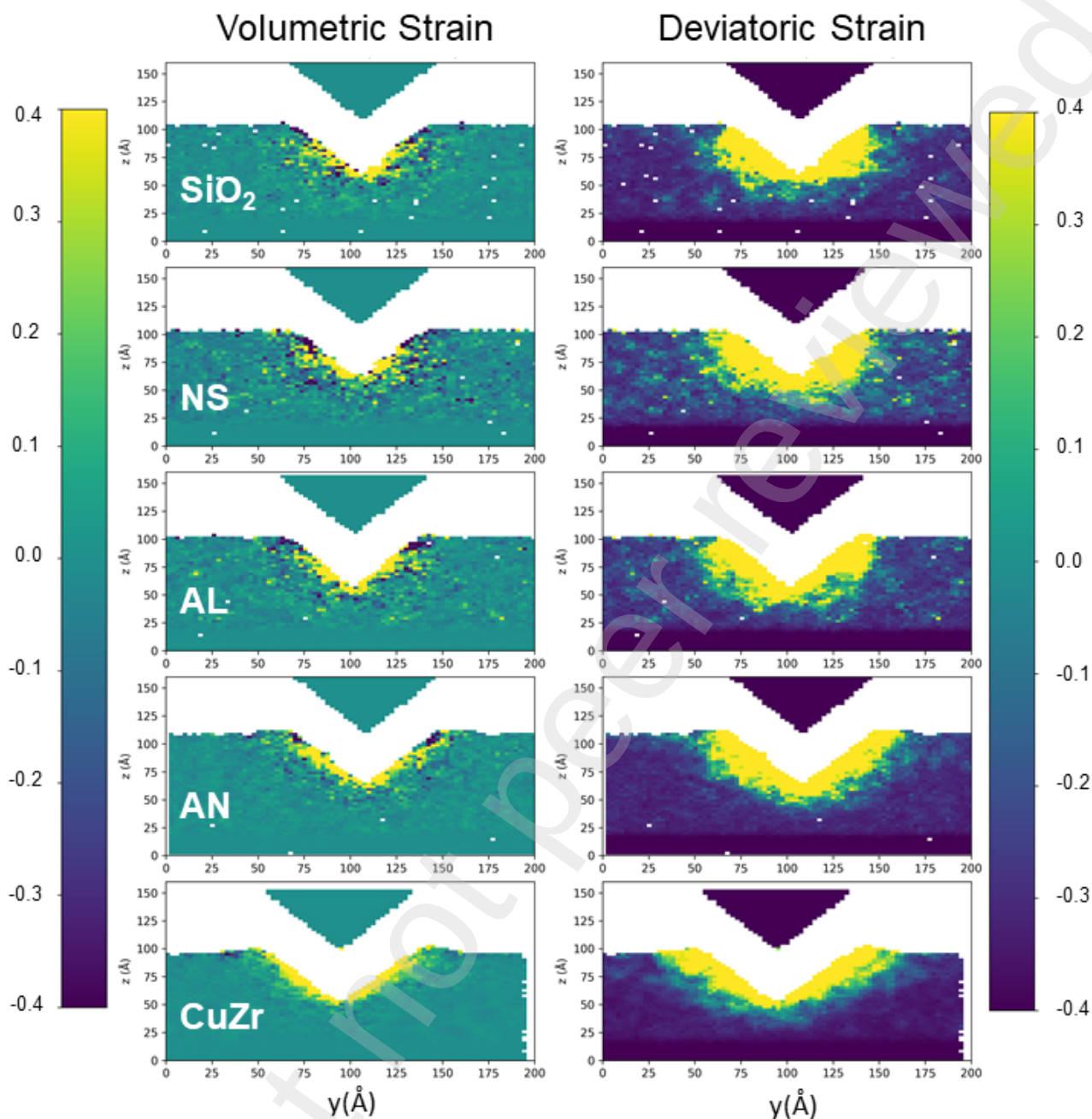
In contrast, vitreous  $\text{SiO}_2$  displays a fundamentally different deformation pattern. Strong densification is localized directly beneath the indenter, while lateral regions show significantly smaller volumetric changes. The corresponding  $D^2_{\min}$  map reveals that non-affine rearrangements are also largely confined below the tip and are much less spatially extended than in the metallic glass. This behavior reflects a densification-dominated mechanism in which plastic deformation is accommodated primarily through network compaction rather than through widespread shear transformations.

The modified silicate glasses (NS, AL, and AN) exhibit mixed behavior. In these systems, the spatial extent of the densified region and the region of elevated  $D^2_{\min}$  are more similar, indicating a coexistence of densification and shear flow. Among the silicate compositions, anorthite (AN) stands out by exhibiting a more pronounced lateral extension of high  $D^2_{\min}$  values and visible pile-up at the surface, making its deformation pattern slightly more similar to that of the metallic glass but with densification map more similar to silicate glasses. This suggests that, within the silicate family, AN accommodates a larger fraction of plastic deformation through shear-like mechanisms, with respect to NS and AL glasses.

The strain maps further clarify these trends. For all glasses, the deviatoric strain fields are more spatially extended than the volumetric strain fields, indicating that locally shear-related deformation spreads more efficiently than volumetric strain. In  $\text{Cu}_{64}\text{Zr}_{36}$ , the deviatoric strain exhibits a wide, radially extended distribution that closely mirrors the  $D^2_{\min}$  maps, including significant shear activity near the free surface. In contrast, for vitreous  $\text{SiO}_2$ , both the deviatoric strain and  $D^2_{\min}$  remain largely confined beneath the indenter, consistent with localized densification-driven plasticity. Differences among the silicate glasses are visible but less pronounced in the strain maps than in the  $D^2_{\min}$  and densification maps, suggesting that the latter provide a more sensitive fingerprint of the underlying deformation mechanism.



**Figure 3.** Atomic-scale densification and non-affine rearrangements after unloading. Two-dimensional maps of residual densification ( $\Delta p/p$ , left column) and non-affine displacement measure ( $D^2_{\min}$ , right column) computed in the *after-unload* configuration for the five investigated glasses: vitreous  $\text{SiO}_2$ , sodium silicate (NS), albite (AL), anorthite (AN), and the CuZr metallic glass (from top to bottom). Each row corresponds to a different glass composition, while the horizontal and vertical axes represent the lateral ( $y$ ) and depth ( $z$ ) directions, respectively.



**Figure 4.** Atomic-scale volumetric and deviatoric strain fields after unloading. Two-dimensional maps of the residual volumetric strain ( $\epsilon_{vol}$ , left column) and equivalent deviatoric strain ( $\epsilon_{dev}$ , right column) evaluated in the *after-unload* configuration for the  $\text{SiO}_2$ , NS, AL, AN, and CuZr glasses, from top to bottom). The axes and layout are identical to those in Figure 3.

### 4.3 Densification-Shear Descriptors.

Based on the residual strain fields shown in Figure 4, a strain threshold of 0.1 was selected to identify plastically active atoms, as this value consistently delineates the region of irreversible deformation beneath the indenter across all investigated glasses while excluding weak elastic fluctuations in the surrounding material.

Figure 5 reports a set of two-dimensional descriptor maps constructed from the strain-based and mechanism-resolving indicators described in section 3.4, including the mean local mode ratio ( $l_1$ ), the fraction of volumetrically dominated atoms ( $l_2$ ), the signed mode index ( $l_3$ ), the mean global mode ratio index ( $l_4$ ), the co-localization index ( $L$ ), and the Spearman correlation coefficient ( $C_s$ ) between densification and non-affine rearrangements ( $D^2_{\min}$ ). All descriptors are evaluated on plastically active atoms in the *after-unload* configuration. In

each panel, the vertical axis represents the extent ratio  $Y = \log_{10} \frac{N(\epsilon^{dev} > \epsilon_{th}^{dev})}{N(|\epsilon^{vol}| > \epsilon_{th}^{vol})}$ , quantifying the relative spatial extent of shear- and densification-dominated deformation.

Across all descriptor maps, the metallic glass  $\text{Cu}_{64}\text{Zr}_{36}$  is clearly separated from the silicate glasses, systematically exhibiting the largest values of  $Y$ . This behaviour reflects the spatially extended nature of shear-driven plasticity in metallic glasses, where deformation is accommodated predominantly through non-affine rearrangements with negligible volumetric compaction.

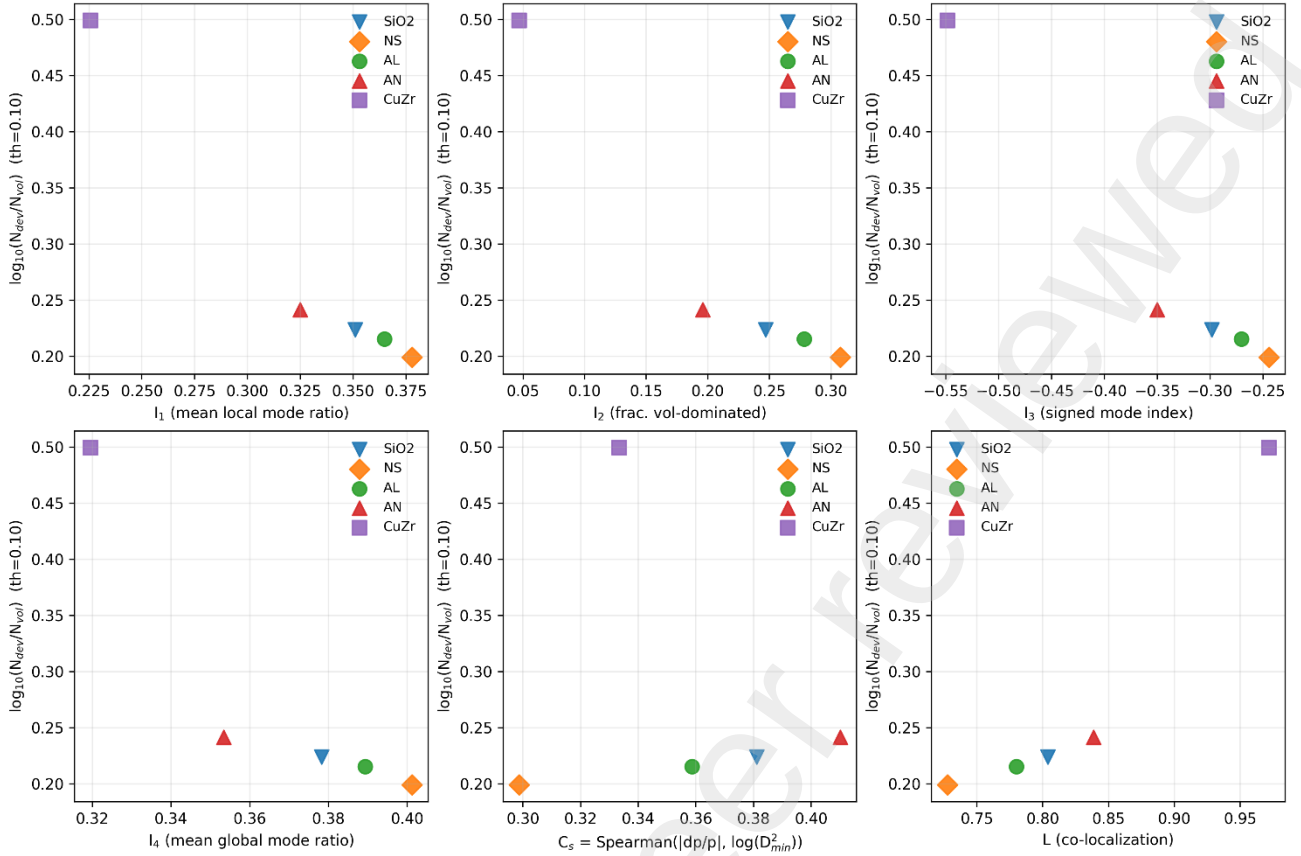
Among the strain-based descriptors,  $l_1$ ,  $l_2$ ,  $l_3$  and  $l_4$  lead to essentially the same qualitative separation of the investigated glasses. These descriptors therefore provide largely redundant information, as they all quantify the relative magnitude of volumetric and deviatoric strain within the plastically active region. While effective in distinguishing metallic from silicate glasses, they do not capture finer differences among oxide compositions. In particular, vitreous  $\text{SiO}_2$  is not clearly separated from NS, AL and AN glasses, despite its well-known propensity for densification under indentation.

The Spearman correlation coefficient  $C_s$  provides complementary information by probing the monotonic coupling between residual densification and non-affine rearrangements within the active zone. Vitreous  $\text{SiO}_2$  exhibits comparatively large  $C_s$ , indicating that atoms undergoing stronger densification also tend to display larger non-affine activity. This is consistent with pressure-assisted network collapse, where permanent densification and topological rearrangements are intrinsically linked in the tetrahedral Si–O network. In contrast, modifier-rich sodium silicate and aluminosilicate glasses display lower  $C_s$  values, reflecting a weaker and more heterogeneous coupling between volumetric compaction and

non-affine rearrangements, likely due to the larger variety of local structural responses enabled by network modifiers and mixed coordination. The introduction of mobile network modifiers (such as Na) and network formers with higher coordination flexibility (Al) progressively weakens the direct coupling between densification and non-affine rearrangements. This reflects the increased ability of these networks to accommodate deformation through a combination of bond bending, sodium mobility leading to local shear flow and coordination expansion leading to local densification, rather than through the highly correlated densification events characteristic of pure silica.

The co-localization index  $L$  captures a different aspect: it quantifies the spatial overlap between densification and shear-related deformation, i.e., whether the same atoms simultaneously participate in volumetric and deviatoric plasticity. Interestingly, the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass exhibits a high  $L$ , comparable to or higher than the silicate glasses. This indicates that, although the metallic glass is shear-dominated in terms of intensity and spatial extent, the atoms participating in plastic flow tend to carry both deviatoric and volumetric components locally. Such behavior is consistent with the dilatant character often associated with shear transformation zones, where shear-driven rearrangements can be accompanied by local free-volume creation/annihilation and therefore by concurrent volumetric changes, even when the net densification averaged over the active zone remains small.

Within the silicate glasses,  $L$  remains relatively high and varies modestly with composition, reflecting the fact that densification and shear are frequently co-activated at the atomic scale in network glasses. Consequently,  $L$  alone does not provide a unique separation between densification-dominated silica and mixed-mechanism silicates, but it complements  $C_s$  by distinguishing “coupling in magnitude” (captured by  $C_s$ ) from “overlap in space” (captured by  $L$ ).



**Figure 5.** Two-dimensional maps of strain-based and mechanism-resolving descriptors evaluated for the five investigated glasses (SiO<sub>2</sub>, NS, AL, AN, and CuZr). All descriptors are computed on plastically active atoms in the *after-unload* configuration, thus reflecting irreversible deformation. In each panel, the vertical axis reports the extent ratio  $Y$ , quantifying the relative spatial extent of shear-dominated versus densification-dominated deformation, while the horizontal axis corresponds to a different descriptor: the Spearman correlation coefficient between densification and non-affine rearrangements ( $C_s$ ), the signed mode index ( $l_3$ ), the mean local mode ratio ( $l_1$ ), the fraction of volumetrically dominated atoms ( $l_2$ ), the co-localization index ( $L$ ), and the mean global mode ratio ( $l_4$ ).

**Figure 6** summarizes the plastic deformation mechanisms of the investigated glasses in the  $\langle \log(D_{min}^2) \rangle - \langle \Delta p/p \rangle$  plane, where both quantities are averaged over plastically active atoms in the *after-unload* configuration. This representation directly contrasts volumetric compaction with non-affine rearrangement activity and reveals a clear separation between shear-dominated and densification-dominated materials.

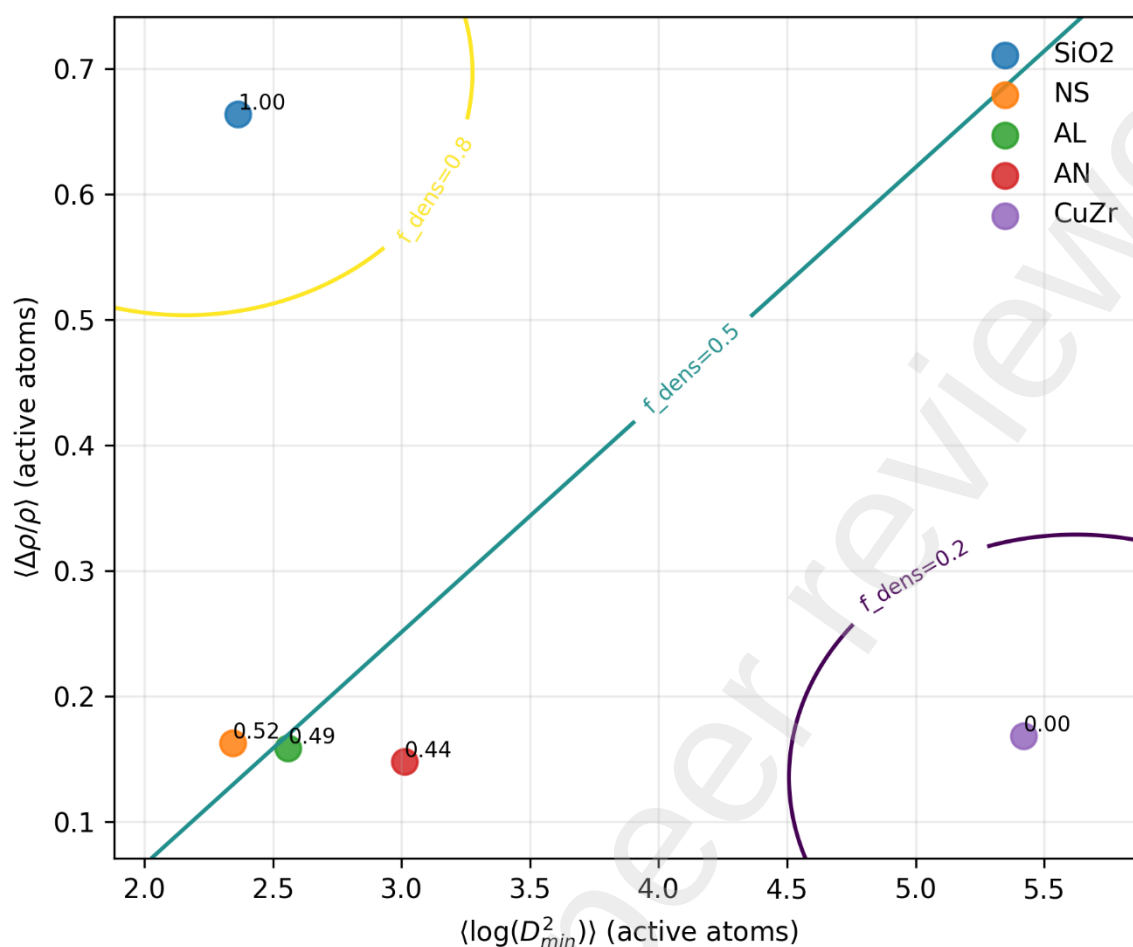
The metallic glass Cu<sub>64</sub>Zr<sub>36</sub> occupies the extreme shear-dominated region of the map, characterized by a very large non-affine activity  $\langle \log(D_{min}^2) \rangle = 5.42$ , combined with a comparatively small residual densification  $\langle \Delta p/p \rangle = 0.17$ . Consistently, the DSF index yields

$f_{dens} = 0$  and  $f_{shear} = 1$ . At the opposite end of the spectrum, vitreous  $\text{SiO}_2$  exhibits strong densification and comparatively weak non-affine activity ( $\langle \log(D_{min}^2) \rangle = 2.36$   $\langle \Delta\rho/\rho \rangle = 0.66$ ) providing a densification fraction  $f_{dens} = 1$ .

The NS, AL, and AN glasses fall between these two end members, occupying intermediate regions of the mechanism map. Their values of  $\langle \log(D_{min}^2) \rangle$  increase progressively from NS to AN, while the densification fraction remains significantly lower than in  $\text{SiO}_2$ . These values indicate that plastic deformation in these oxide glasses results from a mixed mechanism in which densification and shear flow contribute in comparable proportions. The gradual decrease of  $f_{dens}$  from NS to AN reflects an increasing role of shear-driven rearrangements as network connectivity and chemical complexity increase.

The DSF provides a quantitative and continuous measure of the relative contribution of densification and shear flow, interpolating smoothly between the two physically meaningful end members. Unlike strain-only descriptors, which may fail to distinguish silica from other silicates due to the coexistence of volumetric and deviatoric strain, the combined use of residual densification and non-affine rearrangements yields a clear and physically interpretable classification.

Overall, these results demonstrate that the proposed densification–shear fraction index captures the essential differences between metallic and silicate glasses under nanoindentation and provides a robust framework for quantifying mixed deformation mechanisms across chemically diverse materials.



**Figure 6.** Two-dimensional densification–shear mechanism map constructed from Molecular Dynamics nanoindentation simulations. The horizontal axis reports the average non-affine rearrangement intensity,  $\langle \log(D_{min}^2) \rangle$ , while the vertical axis reports the average residual densification,  $\langle \Delta \rho / \rho \rangle$ , both computed over plastically active atoms in the after-unload configuration. Vitreous  $\text{SiO}_2$  and  $\text{Cu}_{64}\text{Zr}_{36}$  define the densification-dominated and shear-dominated end members, respectively. Intermediate silicate and aluminosilicate glasses occupy transitional regions of the map. Curves indicate isolines of constant densification fraction  $f_{dens}$ , quantifying the relative contribution of densification and shear flow to plastic deformation.

#### 4.4 Coordination changes and their relation to densification and shear flow

Figure 7 reports two-dimensional maps of atoms exhibiting an increase in coordination number after unloading ( $\Delta \text{CN} > 1.0$ ), projected onto an  $yz$  slice of the glass slab. For each system, the coordination change is evaluated by comparing the local coordination number in the undeformed reference configuration and in the after-unload configuration, thereby isolating topological changes associated with irreversible deformation. For the silicate

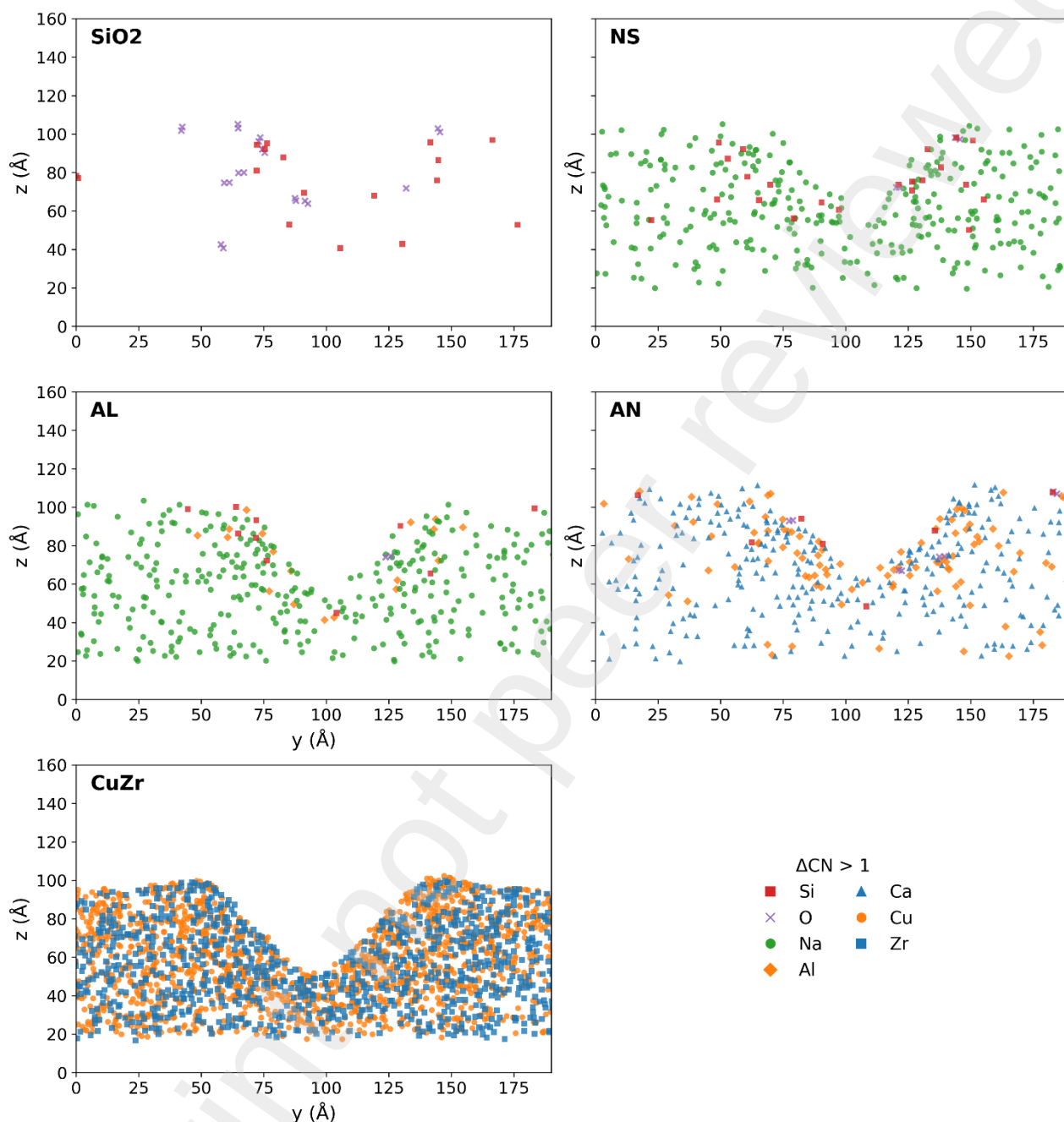
glasses, coordination numbers are computed with respect to oxygen neighbours, while for the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass coordination is evaluated with respect to all neighbouring atoms. A clear trend emerges across the investigated systems: as the contribution of shear flow relative to densification increases, the number of atoms experiencing a significant coordination increase becomes larger and their spatial distribution more extended. In vitreous  $\text{SiO}_2$ , only a limited number of Si atoms exhibit an increase in coordination, and these events are strongly localized beneath the indenter, in the immediate vicinity of the crater formed after unloading. This behaviour is consistent with a densification-dominated mechanism, where coordination changes are rare and confined to regions subjected to the highest hydrostatic pressure.

In contrast, modifier-containing silicate glasses (NS and AL) and the aluminosilicate anorthite glass (AN) show a markedly different behavior. In these systems, atoms undergoing coordination increase are not restricted to the immediate vicinity of the indenter but are distributed over a much broader region of the slab. Notably, sodium atoms—and calcium atoms in AN—exhibit the most widespread coordination increases, extending across the entire yz plane. This reflects the high structural flexibility of modifier species, which readily accommodate shear-driven rearrangements by changing their local coordination environment. In comparison, Si and Al atoms predominantly show coordination increases near the indenter, indicating that network formers are involved mainly in regions experiencing strong local deformation and surface-related shear. Among the two network formers more Al atoms expand their coordination, especially in the AN glass that shows a widespread coordination increase.

The metallic glass, which displays the largest shear contribution among the investigated systems, exhibits coordination increases throughout the entire slab. The affected atoms are broadly distributed but show a higher concentration in the pile-up region surrounding the indent, consistent with shear-flow-dominated plasticity and the absence of significant volumetric compaction. In this case, coordination changes are a direct manifestation of extensive non-affine atomic rearrangements associated with shear transformation zones.

Overall, these coordination maps provide an independent, topological signature of the deformation mechanisms identified through densification, strain, and  $D^2_{\min}$  analyses. The progressive increase in the number and spatial extent of atoms with enhanced coordination, from densification-dominated silica to shear-dominated metallic glass, further supports the interpretation that shear flow promotes widespread coordination changes, particularly in structurally flexible species. This observation complements the descriptor-based analysis

presented above and reinforces the link between atomic-scale topology changes and the relative contributions of densification and shear flow under nanoindentation.



**Figure 7.** Atomic-scale coordination increases after unloading. Two-dimensional maps showing atoms that exhibit an increase in coordination number after unloading ( $\Delta\text{CN} > 0.5$ ) for the five investigated glasses: vitreous  $\text{SiO}_2$ , sodium silicate (NS), albite (AL), anorthite (AN), and the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass. The maps are plotted on a yz slice of the simulation cell with finite thickness along the x direction, centered at the middle of the slab. Coordination changes are evaluated by comparing the undeformed reference configuration with the after-unload configuration, thus highlighting irreversible, topological rearrangements induced by nanoindentation. Cutoffs used are 2.3, 2.1, 2.3, 3.0, 3.0, 3.6 and 3.6 Å for O, Si, Al, Na, Ca, Cu and Zr, respectively.

## Conclusions

In this work, we have introduced a unified atomistic framework to quantify and compare densification and shear flow during MD simulations of nanoindentation tests across chemically distinct glassy systems. By combining local volumetric and deviatoric strain measures, atomic-scale densification, non-affine displacement analysis, and coordination changes, we have provided a comprehensive description of irreversible deformation mechanisms spanning oxide and metallic glasses.

Application of this framework to vitreous  $\text{SiO}_2$ , sodium silicate, albite, anorthite glasses and the  $\text{Cu}_{64}\text{Zr}_{36}$  metallic glass reveals clear and systematic trends. Vitreous silica exhibits strongly localized densification beneath the indenter with limited non-affine activity, consistent with a densification-dominated plastic response. In contrast, the metallic glass displays widespread shear-driven plasticity characterized by extended non-affine rearrangements, pronounced pile-up, and minimal volumetric compaction. Modified silicate and aluminosilicate glasses occupy an intermediate regime, where densification and shear flow coexist, with anorthite showing a stronger shear contribution among the oxide compositions.

We demonstrate that commonly used strain-based descriptors capture important aspects of the deformation intensity but are insufficient, on their own, to provide a transferable quantification of the relative contribution of densification and shear across different glass families. To address this limitation, we have introduced an end-member-based Densification–Shear Fraction (DSF) index, constructed from residual atomic densification and non-affine rearrangement intensity after unloading. Anchored to physically meaningful densification- and shear-dominated reference systems, the DSF provides a continuous, normalized measure of mixed deformation mechanisms that enables direct comparison between oxide and metallic glasses within a single framework.

Independent analysis of coordination number changes further supports this interpretation. The progressive increase in the number and spatial extent of atoms undergoing coordination increases, from densification-dominated silica to shear-dominated metallic glass, highlights the close link between shear flow and topological rearrangements, particularly involving structurally flexible species such as network modifiers and metallic atoms.

Overall, the proposed descriptors establish a robust and transferable atomic-scale methodology for disentangling densification and shear flow under nanoindentation. Beyond the specific systems studied here, this framework provides a general basis for rationalizing composition–structure–mechanism relationships in glasses and opens new avenues for the

interpretation of experimental indentation data and for the design of glass compositions with tailored resistance to contact-induced damage.

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### **Supplementary Information.**

The Supplementary Information reports the full computational details, including the interatomic potentials for oxide and Cu–Zr glasses and the indenter/tip–glass interactions, together with additional atomistic maps of densification, non-affine deformation and strain fields, and the scripts and input data files used to compute the descriptors and generate the figures (on zenodo after publication of the article).

**Use of Artificial Intelligence.** Artificial intelligence tools were used in this work to improve the clarity and quality of the written language and to assist with code development and debugging. All scientific interpretations, data analysis, and conclusions are the sole responsibility of the authors.

### **References**

- [1] V. Keryvin, L. Charleux, R. Hin, J.-P. Guin, J.-C. Sangleboeuf, Mechanical behaviour of fully densified silica glass under Vickers indentation, *Acta Mater.* 129 (2017) 492–499. <https://doi.org/10.1016/j.actamat.2017.03.008>.
- [2] J. Chen, S.J. Bull, Multi-cycling nanoindentation study on thin optical coatings on glass, *J. Phys. Appl. Phys.* 41 (2008) 074009. <https://doi.org/10.1088/0022-3727/41/7/074009>.
- [3] K.E. Avila, S. Küchemann, H.M. Urbassek, Structure and size of the plastic zone formed during nanoindentation of a metallic glass, *J. Non-Cryst. Solids* 523 (2019) 119593. <https://doi.org/10.1016/j.jnoncrsol.2019.119593>.
- [4] X. Li, B. Bhushan, A review of nanoindentation continuous stiffness measurement technique and its applications, *Mater. Charact.* 48 (2002) 11–36. [https://doi.org/10.1016/S1044-5803\(02\)00192-4](https://doi.org/10.1016/S1044-5803(02)00192-4).
- [5] T. Burgess, M. Ferry, Nanoindentation of metallic glasses, *Mater. Today* 12 (2009) 24–32. [https://doi.org/10.1016/S1369-7021\(09\)70039-2](https://doi.org/10.1016/S1369-7021(09)70039-2).

- [6] P.M. Rice, R.E. Stoller, Correlation of Nanoindentation and Conventional Mechanical Property Measurements, *MRS Online Proc. Libr.* 649 (2011) 711. <https://doi.org/10.1557/PROC-649-Q7.11>.
- [7] T. Rouxel, J. Jang, U. Ramamurty, Indentation of glasses, *Prog. Mater. Sci.* 121 (2021) 100834. <https://doi.org/10.1016/j.pmatsci.2021.100834>.
- [8] T. Rouxel, H. Ji, J.P. Guin, F. Augereau, B. Rufflé, Indentation deformation mechanism in glass: Densification versus shear flow, *J. Appl. Phys.* 107 (2010) 094903. <https://doi.org/10.1063/1.3407559>.
- [9] H. Ji, V. Keryvin, T. Rouxel, T. Hammouda, Densification of window glass under very high pressure and its relevance to Vickers indentation, *Scr. Mater.* 55 (2006) 1159–1162. <https://doi.org/10.1016/j.scriptamat.2006.08.038>.
- [10] E. Barthel, V. Keryvin, G. Rosales-Sosa, G. Kermouche, Indentation cracking in silicate glasses is directed by shear flow, not by densification, *Acta Mater.* 194 (2020) 473–481. <https://doi.org/10.1016/j.actamat.2020.05.011>.
- [11] G. Kermouche, E. Barthel, D. Vandembroucq, Ph. Dubujet, Mechanical modelling of indentation-induced densification in amorphous silica, *Acta Mater.* 56 (2008) 3222–3228. <https://doi.org/10.1016/j.actamat.2008.03.010>.
- [12] C. Hermansen, J. Matsuoka, S. Yoshida, H. Yamazaki, Y. Kato, Y.Z. Yue, Densification and plastic deformation under microindentation in silicate glasses and the relation to hardness and crack resistance, *J. Non-Cryst. Solids* 364 (2013) 40–43. <https://doi.org/10.1016/j.jnoncrysol.2012.12.047>.
- [13] M.N. Svenson, T.K. Bechgaard, S.D. Fuglsang, R.H. Pedersen, A.Ø. Tjell, M.B. Østergaard, R.E. Youngman, J.C. Mauro, S.J. Rzoska, M. Bockowski, M.M. Smedskjaer, Composition-Structure-Property Relations of Compressed Borosilicate Glasses, *Phys. Rev. Appl.* 2 (2014) 024006. <https://doi.org/10.1103/PhysRevApplied.2.024006>.
- [14] S. Kasimuthumaniyan, S. Sahoo, M.M. Smedskjaer, N.M.A. Krishnan, N.N. Gosvami, Quantifying the densification and shear flow under indentation deformation in borosilicate glasses, *Int. J. Appl. Glass Sci.* 13 (2022) 526–538. <https://doi.org/10.1111/ijag.16598>.
- [15] T. Rouxel, H. Ji, J.P. Guin, F. Augereau, B. Rufflé, Indentation deformation mechanism in glass: Densification versus shear flow, *J. Appl. Phys.* 107 (2010) 094903. <https://doi.org/10.1063/1.3407559>.

- [16] A. Perriot, D. Vandembroucq, E. Barthel, V. Martinez, L. Grosvalet, Ch. Martinet, B. Champagnon, Raman Microspectroscopic Characterization of Amorphous Silica Plastic Behavior, *J. Am. Ceram. Soc.* 89 (2006) 596–601. <https://doi.org/10.1111/j.1551-2916.2005.00747.x>.
- [17] D. Vandembroucq, T. Deschamps, C. Coussa, A. Perriot, E. Barthel, B. Champagnon, C. Martinet, Density hardening plasticity and mechanical ageing of silica glass under pressure: a Raman spectroscopic study, *J. Phys. Condens. Matter* 20 (2008) 485221. <https://doi.org/10.1088/0953-8984/20/48/485221>.
- [18] T. Rouxel, Elastic Properties and Short-to Medium-Range Order in Glasses, *J. Am. Ceram. Soc.* 90 (2007) 3019–3039. <https://doi.org/10.1111/j.1551-2916.2007.01945.x>.
- [19] P. Sellappan, T. Rouxel, F. Celarie, E. Becker, P. Houizot, R. Conradt, Composition dependence of indentation deformation and indentation cracking in glass, *Acta Mater.* 61 (2013) 5949–5965. <https://doi.org/10.1016/j.actamat.2013.06.034>.
- [20] A.S. Argon, Plastic deformation in metallic glasses, *Acta Metall.* 27 (1979) 47–58. [https://doi.org/10.1016/0001-6160\(79\)90055-5](https://doi.org/10.1016/0001-6160(79)90055-5).
- [21] M.L. Falk, J.S. Langer, Dynamics of viscoplastic deformation in amorphous solids, *Phys. Rev. E* 57 (1998) 7192–7205. <https://doi.org/10.1103/PhysRevE.57.7192>.
- [22] C.A. Schuh, T.C. Hufnagel, U. Ramamurty, Mechanical behavior of amorphous alloys, *Acta Mater.* 55 (2007) 4067–4109. <https://doi.org/10.1016/j.actamat.2007.01.052>.
- [23] D.A. Kilymis, J.M. Delaye, Nanoindentation of pristine and disordered silica: Molecular Dynamics simulations, *J. Non-Cryst. Solids* 382 (2013) 87–94. <https://doi.org/10.1016/j.jnoncrysol.2013.10.013>.
- [24] N. Amigo, F. Valencia, Mechanical and structural assessment of CuZr metallic glasses rejuvenated by thermal-pressure treatments, *Comput. Mater. Sci.* 198 (2021) 110681. <https://doi.org/10.1016/j.commatsci.2021.110681>.
- [25] O. Adjaoud, K. Albe, Nanoindentation of Nanoglasses Tested by Molecular Dynamics Simulations: Influence of Structural Relaxation and Chemical Segregation on the Mechanical Response, *Front. Mater.* 8 (2021). <https://doi.org/10.3389/fmats.2021.664220>.
- [26] Z. Yan, R. Lu, L. Ding, L. Wang, Z. Zhang, Distinct deformation mechanisms of silicate glasses under nanoindentation: The critical role of structure, *J. Appl. Phys.* 136 (2024) 245101. <https://doi.org/10.1063/5.0244335>.
- [27] H. Liu, B. Deng, S. Sundararaman, Y. Shi, L. Huang, Understanding the response of aluminosilicate and aluminoborate glasses to sharp contact loading using molecular

- dynamics simulation, *J. Appl. Phys.* 128 (2020) 035106. <https://doi.org/10.1063/5.0013555>.
- [28] L. Ding, R. Lu, L. Wang, Q. Zheng, J.C. Mauro, Z. Zhang, Nanoindentation-induced evolution of atomic-level properties in silicate glass: Insights from molecular dynamics simulations, *J. Am. Ceram. Soc.* 107 (2024) 1448–1458. <https://doi.org/10.1111/jace.19230>.
- [29] A. Pedone, Properties Calculations of Silica-Based Glasses by Atomistic Simulations Techniques: A Review, *J. Phys. Chem. C* 113 (2009) 20773–20784. <https://doi.org/10.1021/jp9071263>.
- [30] A.P. Thompson, H.M. Aktulga, R. Berger, D.S. Bolintineanu, W.M. Brown, P.S. Crozier, P.J. in 't Veld, A. Kohlmeyer, S.G. Moore, T.D. Nguyen, R. Shan, M.J. Stevens, J. Tranchida, C. Trott, S.J. Plimpton, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, *Comput. Phys. Commun.* 271 (2022) 108171. <https://doi.org/10.1016/j.cpc.2021.108171>.
- [31] G.J. Martyna, M.L. Klein, M. Tuckerman, Nosé–Hoover chains: The canonical ensemble via continuous dynamics, *J. Chem. Phys.* 97 (1992) 2635–2643. <https://doi.org/10.1063/1.463940>.
- [32] M. Bertani, M.C. Menziani, A. Pedone, Improved empirical force field for multicomponent oxide glasses and crystals, *Phys. Rev. Mater.* 5 (2021) 045602. <https://doi.org/10.1103/PhysRevMaterials.5.045602>.
- [33] A. Pallini, B. Ziebarth, W. Mannstadt, A. Pedone, Pressure-induced structural variations and mechanical behavior of silicate glasses: Role of aluminum and sodium, *J. Non-Cryst. Solids* 647 (2025) 123267. <https://doi.org/10.1016/j.jnoncrysol.2024.123267>.
- [34] A. Pallini, M. Bertani, D. Rustichelli, B. Ziebarth, W. Mannstadt, A. Pedone, Comparison of five empirical potential models for aluminosilicate systems: Albite and anorthite as test cases, *J. Non-Cryst. Solids* 615 (2023) 122426. <https://doi.org/10.1016/j.jnoncrysol.2023.122426>.
- [35] M.I. Mendelev, D.J. Sordet, M.J. Kramer, Using atomistic computer simulations to analyze x-ray diffraction data from metallic glasses, *J. Appl. Phys.* 102 (2007) 043501. <https://doi.org/10.1063/1.2769157>.
- [36] M.I. Mendelev, M.J. Kramer, R.T. Ott, D.J. Sordet, D. Yagodin, P. Popel, Development of suitable interatomic potentials for simulation of liquid and amorphous Cu–Zr alloys, *Philos. Mag.* 89 (2009) 967–987. <https://doi.org/10.1080/14786430902832773>.

- [37] H.W. Sheng, W.K. Luo, F.M. Alamgir, J.M. Bai, E. Ma, Atomic packing and short-to-medium-range order in metallic glasses, *Nature* 439 (2006) 419–425. <https://doi.org/10.1038/nature04421>.
- [38] J. Ding, S. Patinet, M.L. Falk, Y. Cheng, E. Ma, Soft spots and their structural signature in a metallic glass, *Proc. Natl. Acad. Sci.* 111 (2014) 14052–14056. <https://doi.org/10.1073/pnas.1412095111>.
- [39] C. Tang, P. Harrowell, Anomalously slow crystal growth of the glass-forming alloy CuZr, *Nat. Mater.* 12 (2013) 507–511. <https://doi.org/10.1038/nmat3631>.
- [40] W.C. Oliver, G.M. Pharr, Measurement of hardness and elastic modulus by instrumented indentation: Advances in understanding and refinements to methodology, *J. Mater. Res.* 19 (2004) 3–20. <https://doi.org/10.1557/jmr.2004.19.1.3>.
- [41] J. Tersoff, Empirical Interatomic Potential for Carbon, with Applications to Amorphous Carbon, *Phys. Rev. Lett.* 61 (1988) 2879–2882. <https://doi.org/10.1103/PhysRevLett.61.2879>.
- [42] W.C. Oliver, G.M. Pharr, An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments, *J. Mater. Res.* 7 (1992) 1564–1583. <https://doi.org/10.1557/JMR.1992.1564>.
- [43] J.A. Zimmerman, D.J. Bammann, H. Gao, Deformation gradients for continuum mechanical analysis of atomistic simulations, *Int. J. Solids Struct.* 46 (2009) 238–253. <https://doi.org/10.1016/j.ijsolstr.2008.08.036>.
- [44] K. Januchta, T. To, M.S. Bødker, T. Rouxel, M.M. Smedskjaer, Elasticity, hardness, and fracture toughness of sodium aluminoborosilicate glasses, *J. Am. Ceram. Soc.* 102 (2019) 4520–4537. <https://doi.org/10.1111/jace.16304>.
- [45] F.J.P. Sousa, M. Dal Bó, P.O. Guglielmi, R. Janssen, D. Hotza, Characterization of Young's modulus and fracture toughness of albite glass by different techniques, *Ceram. Int.* 40 (2014) 10893–10899. <https://doi.org/10.1016/j.ceramint.2014.03.085>.
- [46] D. Xu, B. Lohwongwatana, G. Duan, W.L. Johnson, C. Garland, Bulk metallic glass formation in binary Cu-rich alloy series – Cu<sub>100-x</sub>Zr<sub>x</sub> (x=34, 36, 38.2, 40 at.%) and mechanical properties of bulk Cu<sub>64</sub>Zr<sub>36</sub> glass, *Acta Mater.* 52 (2004) 2621–2624. <https://doi.org/10.1016/j.actamat.2004.02.009>.
- [47] J.E. Neely, J.D. Mackenzie, Hardness and low-temperature deformation of silica glass, *J. Mater. Sci.* 3 (1968) 603–609. <https://doi.org/10.1007/BF00757906>.
- [48] B. Lawn, *Fracture of Brittle Solids*, 2nd ed., Cambridge University Press, Cambridge, 1993. <https://doi.org/10.1017/CBO9780511623127>.

- [49] M. Kazembeyki, K. Yang, J.C. Mauro, M.M. Smedskjaer, M. Bauchy, C.G. Hoover, Decoupling of indentation modulus and hardness in silicate glasses: Evidence of a shear- to densification-dominated transition, *J. Non-Cryst. Solids* 553 (2021) 120518. <https://doi.org/10.1016/j.jnoncrysol.2020.120518>.
- [50] D.R. Tadjiev, R.J. Hand, Surface hydration and nanoindentation of silicate glasses, *J. Non-Cryst. Solids* 356 (2010) 102–108. <https://doi.org/10.1016/j.jnoncrysol.2009.10.005>.